

Rendering Reality: Measuring Time Efficiency in Virtual Human Creation

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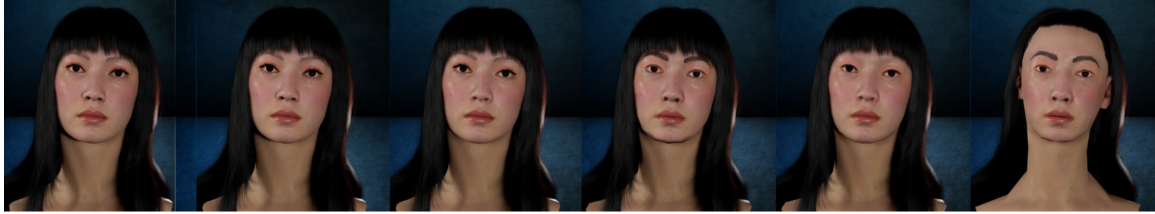


Figure 1: Virtual Human representation used in experiment, in descending Level of Detail.

ABSTRACT

Recent advancements in real-time rendering technology have significantly improved the detail of virtual humans. Higher resolution textures, physically-based hair and skin materials, and other innovations have increased their likeness. However, this has also raised render time and complexity. While hardware improvements alleviate some issues, limited access to high-performance hardware restricts creators on lower-performance systems from using advanced rendering techniques. This study explores the relationship between render times at different levels of detail and the perceived realism of avatars. By comparing these levels, we establish a relative realism scale and identify suitable levels of detail for specific virtual experiences.

Index Terms: Virtual Humans, Perceived Realism, Render Style

1 INTRODUCTION

The fidelity of virtual humans or avatars elicits diverse emotional responses during interaction. The most documented is the eeriness or fear associated with almost-realistic human representations, termed the “uncanny valley” [3]. This describes the emotional trough between a real human and its representation, which intensifies when these representations are animated. Uncanny effects are also seen in virtual humans with slightly altered or abnormal appearances [4]. Features such as large eyes or a wide mouth evoke similar emotional responses as less realistic avatars. MacDorman and Chattopadhyay [2] examined the uncanny valley by varying facial features (eyes, mouth, skin) to create different realism levels, finding that uncanny effects derive from transitioning from real to computer-modeled avatars. Realism inconsistencies make anthropomorphic characters seem unfamiliar despite physical similarity to real people. Viewer sensitivity to these inconsistencies drives characters into the uncanny valley. Deviations in facial features and animations from expected realism elicit strong negative emotions, intensifying the effect. Our study explores the relationship between render times of virtual humans and their levels of detail.

2 METHODS

2.1 Materials

The avatars were created using MetaHuman Creator for Unreal Engine (UE) version 5.4.2. Rendering techniques included an albedo

map [5] for base color and light reflectivity, a specular map for reflections on stretched skin areas, and a scatter map to model light dispersion in thin skin regions like the nose, ears, and lips. Several normal maps were also included to provide more detailed skin textures, which were blended with their respective facial expressions to add detail. Lastly, roughness maps were generated from the respective normal maps for reflective areas of the skin. All maps were rendered in 4K resolution. Hair rendering followed methods by d'Eon, Marschner, and Hanika [1], improving noise reduction, computational efficiency, and shader parameter variation. MetaHuman provides eight different levels of detail for an avatar, of which the six highest levels of detail were used in this study to maintain a manageable survey question size. MetaHuman's dynamic level of detail system provides a varied degradation of the avatar, reducing the total model vertices, joints, and skin influences¹. Additionally, MetaHuman degrades hair rendering for less detailed virtual humans [1]. An idle animation was applied for assessing realism through motion [3]. Captured with an Apple iPhone 13 Pro via the LiveLinkFace app, animation data was exported into a UE 5.4.2 scene featuring a three-point lighting setup, a perspective camera, and post-processing for anti-aliasing and denoising to improve clarity and reduce artifacts.

Scenes were replicated for all levels of detail, with all objects, except the level of detail kept identical. Each scene was rendered using the Movie Render Pipeline for UE 5.4.2. in the UE Sequencer. The sequence was rendered at 30 FPS for 2 seconds at a resolution of 1920 and then down sampled to 960. Image sequences were then further downsampled to a width of 450 in their representation on the survey. Zooming in on images inside the browser would have shown higher-resolution image sequences. The render time was recorded from the Unreal Engine window. The computer used for rendering had 32 GB of memory, an Nvidia GeForce RTX 3080Ti GPU, and an Intel Core i9-12900K CPU.

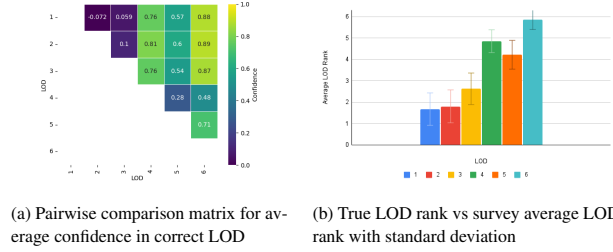
2.2 Participants, Procedures and Measures

A total of 79 participants completed an anonymous survey. The survey was distributed to participants through direct messaging and online forums. The survey was created using the Qualtrics survey platform and consisted of two sections. The first section contained 15 side-by-side comparisons of all possible combinations of two different levels of detail. Participants were presented with two videos and asked to evaluate which one appeared more realistic. The responses were captured using a 7-point Likert scale. The scale included answer choices for either the first or second avatar, with three levels of confidence for each choice, and an option for no discernible difference. Responses ranged from -1 (preference for less detailed) to 1 (preference for more detailed), with an option

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for no discernible difference. The second section asked participants to rank the avatars based on their perceived realism. Participants were shown all levels of detail and asked to provide a ranking from 1 to 7, with 1 being the most realistic and 7 being the least realistic.

3 RESULTS

A one-way analysis of variance was conducted for all pairwise comparison data ($F_{14,1170} = 89.54, p < .001$). Likert scale values were averaged for all survey responses to create a triangular pairwise comparison matrix (figure a). All values were then averaged, resulting in a total average value ($M = 0.55, SD = 0.12$). While there were not statistically significant differences between column averages (total average standard deviation overlaps many of the values); there were significant differences in confidence between high detail avatars (LOD 1-3) and low detail avatars (LOD 4-6) (figure b). Uncertainty was the lowest when comparing high-detail avatars to other high-detail avatars, while low-detail avatars could be differentiated from other low-detail avatars with a significantly higher degree of confidence.

A logarithmic regression analysis was conducted to find the relationship between the level of detail and the decrease in render time (Table 1) ($R^2 = 0.9817, y = 0.16 + 4.31 \cdot \ln(x)$). This indicates a strong correlation between render time and detail, with render time showing diminishing returns as the level of detail increases. The most time-efficient option was the third-highest level of detail as it had the highest render time save for its perceived realism.

A Spearman rank correlation coefficient and Pearson correlation coefficient were calculated between the true rank and average rank ($\rho = 0.99, r = 0.94$), indicating a strong linear relationship between perceived rank and true rank.

4 DISCUSSION AND CONCLUSION

An important decision in this project was not to use objective metric values when rating avatar realism. Objective metrics had the issue of users having different perspectives for realism. Using rank data and pairwise comparison data provides information on the relative differences between level of detail and perceived realism that were more important to finding a relationship between the two.

The results indicate that there exists a logarithmic relationship between level of detail and perceived realism through rank. Since each specific level of detail had a corresponding render time, there was a logarithmic relationship with a high correlation coefficient ($R^2 = 0.98$) between level of detail and render time difference (Table 1). As such, we can conclude that the biggest time saves come from the second and third level of detail.

Additionally, with true rank vs average rank data not deviating as much for high detail avatars as opposed to lower detail avatars (figure b), and significant differences of average confidence in the pairwise comparison matrix (figure a), reveals that users were uncertain in rating more detailed virtual humans than they were in less detailed ones. Sufficient detail is all that is needed to convince the viewer of realism.

By finding a confidence metric in pairwise comparisons, and correlating that with differences in average rank, the level of detail

Table 1: Metrics of average confidence, render time (seconds), and difference in time from LOD 0

LOD	1	2	3	4	5	6
Confidence	0.44	0.46	0.46	0.52	0.44	0.76
Render Time	15.43	12.42	10.20	8.74	8.32	8.10
Difference	0	3.01	5.23	6.69	7.11	7.33

in virtual humans matters for a significant level of perceived realism. However, more detailed representations of virtual humans have diminishing returns in perceived realism with the biggest tax on render time. Confidence also decreases with high levels of detail. The study finds the most time efficient choice in the Unreal Engine MetaHuman system would be the third level of detail as it shares the low confidence between the other detailed virtual humans, while still keeping a statistically significant high rank in realism.

This study has several limitations to consider when generalizing its findings. First, the virtual human used had a specific demographic, limiting applicability to avatars with different demographics. Another limitation was that videos were observed through screens, however observing the avatar in a VR/XR context may have changed the total perceived realism [6]. Second, the survey did not collect demographic data from participants, which could have revealed how different groups perceive realism. Third, the renderings were downscaled twice: once to meet file size requirements and again to fit side-by-side images in the survey. This downscaling may have removed critical details necessary to differentiate levels of detail. Fourth, using more levels of detail could have provided deeper insights. While distinct differences were observed between high and low levels, intermediate levels might reveal a more nuanced relationship. For example, participants showed a preference for LOD 5 over LOD 4, suggesting future studies could explore specific facial features contributing to perceived realism and why lower detail levels introduce greater variability. Next, the avatars were rendered using Unreal Engine Rendering pipeline on specific computer hardware, different systems may yield different results. Lastly, hair rendering was problematic at lower detail levels. As Unreal Engine's MetaHuman system automatically adjusts hair rendering with detail changes, future studies could control hair rendering independently to assess its role in perceived realism.

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